

1. Overview

This document defines the MQTT communication interface between the Switch device and the Management Platform.

2. MQTT Topics

The following MQTT topics are defined:

switch/<sn>/first_seen discovery	(Switch → Platform) Device onboarding / discovery
switch/<sn>/request	(Platform → Switch) Send command requests
switch/<sn>/response	(Switch → Platform) Command execution results
switch/<sn>/event alarms	(Switch → Platform) Device notifications / alarms
switch/<sn>/status	(Switch → Platform) Device state updates
switch/<sn>/telemetry	(Switch → Platform) Periodic statistics
switch/<sn>/file_transfer transfer	(Platform → Switch) Firmware or configuration transfer
Switch/<sn>/process progress of	(Switch → Platform)Report the execution file_transfer
switch/all/request	(Platform → All Switches) Broadcast command

3. Status Code Definition

0 = Success

1 = Invalid command

2 = Invalid parameters

3 = Command execution failed

4 = Device busy

5 = Internal device error

4. Device Onboarding (First Seen)

When a switch connects to the MQTT broker for the first time or boots up, it should publish device information.

Topic:

switch /<sn>/first_seen

Purpose:

- Allow automatic device discovery
- Store device information in platform database
- Enable onboarding workflow in UI

Example First Seen Payload

```
{  
  "serial_number": "SN123456789",  
  "mac_address": "00:11:22:33:44:55",  
  "model": "TR-SW-24G",  
  "hardware_version": "1.0",  
  "software_version": "2.3.1",  
  "ip_address": "192.168.1.10/24",  
  "timestamp": 1710000000  
}
```

Note:

"ip_address" - represents the IPv4 address/mask of the connection to the broker.

5. Command Request Format / Command Response

Format



AC5 Switch
CLI-MQTT.docx

Reference documents

Command Request Topic

switch /<sn>/request

Payload structure:

```
{  
  "request_id": "string",  
  "command": "string",  
  "params": {  
  }  
}
```

Field Description:

request_id: Unique request identifier

command: Command name

params: Command parameters

Example Request

```
{  
  "request_id": "req_1001",  
  "command": "show version",  
  "params": {  
  }  
}
```

Command Response Topic

switch /<sn>/response

Payload structure:

```
{  
  "request_id": "string",  
  "status": 0,  
  "message": " string ",  
  "data": "string"  
}
```

Example Response

```
{  
  "request_id": "req_1001",  
  "status": 0,  
  "message": " Success",
```

```
"data": " Software Version: xxxx \n Release Date: 2026-02-26 22:41:27\n      System MAC Address: 70:98:cd:bb:98:50\n      PID: xxxx , Hardware\n      Version: 1.0, Serial Number: xxxx \n Vendor oid: xxxx \n "
```

}

6. Broadcast Command

Topic:

switch/all/request

Purpose:

Send a command to all switches simultaneously.

All switches should subscribe to this topic and execute the command when received.

The " command " is consistent with the description in the Command Request Format.

7. Event Notification

Topic:

switch /<sn> /event

Events represent asynchronous notifications generated by the switch.

Examples:

- port down
- port up

Example Event Payload

```
{  
  "event": " port_ up ",  
  "timestamp": 1710000000,  
  "data": {  
    "port": "eth1"  
  }  
}
```

```
{  
  "event": "port_down",  
  "timestamp": 1710000000,  
  "data": {  
    "port": "eth1"  
  }  
}
```

8. Device Status Reporting

Topic:

switch /<sn>/status

Used to report device lifecycle state such as online ~~/offline or rebooting~~.

Example Status Payload

```
{
```

```
"status": "online",  
"uptime": 123456,  
"software_version": "2.3.1"  
}
```

Note:

1. Offline implementation relies on the MQTT LWT (Last Will and Testament) mechanism, with an uptime of zero.
2. Rebooting cannot guarantee that the notification message will be delivered.

Telemetry Reporting

Topic:

switch /<sn>/telemetry

The switch should periodically publish telemetry information (recommended interval: 30 seconds). We're considering whether a 30 seconds interval is too short, as it could put continuous pressure on the platform. Therefore, we've added a configurable command: "mqtt client report-interval <10-65535>". What should the default value be?

Telemetry includes system metrics and interface metrics.

Telemetry System Metrics

Required system telemetry fields:

- cpu_usage

- memory_usage
- temperature
- serial_number
- software_version

9. Telemetry Interface Metrics

For each interface (port), the following information should be provided:

- port
- status
- mac_address
- ip_address
- rx_bytes
- tx_bytes
- rx_packets
- tx_packets

Example Telemetry Payload

```
{  
  "timestamp": 1710000000,  
  "cpu_usage": 15,  
  "memory_usage": 40,  
  "Temperature": 45  
  "serial_number": "SN123456789",  
  "software_version": "2.3.1",  
  "interfaces": [  
    {
```



```
"port": "eth0",  
"status": "up",  
"mac_address": "AA:BB:CC:DD:EE:01",  
"ip_address": "192.168.1.10/24",  
"rx_bytes": 123456,  
"tx_bytes": 654321 ,  
"rx_packets": 12345678,  
"tx_packets": 87654321  
}  
]  
}
```

Note:

"cpu_usage": 15 - indicates that CPU usage is 15%.

"memory_usage": 40 - indicates that memory usage is 40%

JSON uses the double type as the numeric data type, while rx_bytes, tx_bytes, rx_packets, and tx_packets are 64-bit unsigned integers, which may lead to out-of-bounds errors. Furthermore, these statistics are only displayed for physical ports; they are not displayed for SVI ports.

10. File Transfer

Topic:

switch /<sn> /file_transfer

Used to initiate firmware updates, configuration downloads, or other file-based operations.

The device should download the file from the provided URL and verify checksum before applying it.

Algorithm supports :

MD5/SHA1/SHA256/SHA512

file_type support:

Firmware : Firmware upgrade

Backup_configuration: Backup configuration

Restore_configuration: Change configuration

Restore_certificate: Change certificate

Restore_privateKey: Change the key

Restore_ca: Change the ca

Note:

Firmware and Restore_configuration require a device reboot to take effect.

Restore_ca, Restore_certificate and Restore_privateKey require MQTT to be re-enabled.

Example File Transfer Payload

```
{  
  "request_id": "file_001",  
  "file_type": "firmware",  
  "file_url": "https://server/files/fw_v2.1.bin",  
  "algorithm": "MD5",  
  "checksum": "9f3a7c2d",  
  "version": "2.1"  
}
```

Note:

During file operations, it is necessary to prevent restart operations.

Do we need a function to return the progress, indicating that the operation is still ongoing?

To address this, we've added the `switch/<sn>/process` topic to provide real-time updates on file execution status, such as `Transferring/Verifying/Applying`. Furthermore, `Transferring/Applying` will continue to update `process` after a `report-interval`.

Example File Process Payload

```
{  
  "request_id": "file_001",  
  "status": 0,  
  "message": "Success",  
  "data": {  
    "state": "Verifying"  
  }  
}
```

Example success Response

file_type : Firmware / Restore_configuration / Restore_certificate /
Restore_privateKey

```
{  
  "request_id": "file_001",  
  "status": 0,  
  "message": " Success",  
  "data": {
```

```
}  
}  
  
file_type : Backup_configuration  
  
{  
  "request_id": "file_001",  
  "status": 0,  
  "message": " Success",  
  "data": {  
    "algorithm": "MD5",  
    "checksum" : "9f3a7c2d"  
  }  
}
```

Example failed response

```
timeout:  
  
{  
  "request_id": "file_001",  
  "status": 3 ,  
  "message": "Command execution failed",  
  "data": {  
  }  
}
```

Firmware error or insufficient memory:

```
{  
  "request_id": "file_001",  
  "status": 5 ,  
  "message": "Internal device error",  
  "data": {  
  }  
}
```

Checksum error:

```
{  
  "request_id": "file_001",  
  "status": 3 ,  
  "message": "Invalid parameters",  
  "data": {  
  }  
}
```

11. Necessary configuration

Note:

1. Please synchronize your clock first to prevent issues arising outside the certificate's effective timeframe.
2. "commander" is the alias we're using for now; you can provide a better one later.
3. The upgraded certificate will be lost after a reboot; we will make it permanent in a future version.

SWITCH#upgrade mqtt-ca URL

SWITCH#upgrade mqtt-client-cert URL

SWITCH#upgrade mqtt-client-key URL

SWITCH(config)#mqtt client server URL
SWITCH(config)#mqtt client scheme commander
SWITCH(config)#mqtt client enable